

Functional genetic report · longevity

Heart Rhythm and Arrhythmias

Sérgio F · Report date: 03-10-2026

As the genes regulate the natural heart pacemaker, vagal brake (measured by heart rate variability), electrical conduction and risk of arrhythmias. It complements the sections of atrial fibrillation and QT of the advanced cardiovascular report. Only genes (chip + local imputation + polygenic scores).

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Summary

18

SNPs analyzed

17

in the chip

0

imputed ($DR2 \geq 0.8$)

1

SNP not found in the data

10

Polygenic scores

The rhythm is born in the sinus node, where the HCN4 channel generates the "funny" current that triggers each beat; the vagus nerve stops this rhythm through the Muscarinic M2 receptor, the proteins G (GNG11) and its regulator RGS6, and the channels GIRK (KCNJ5). Heart rate variability (HRV, for example RMSSD) measures this vagal brake: more variability is associated with better cardiovascular health and less mortality. The conduction depends on sodium channels (SCN5A, SCN10A) and transcription factors (TBX5, HEY2).

In summary: genetically low resting heart rate (80 score variants at the 4th percentile) and favourable conduction and Brugada (SCN10A without broad QRS alleles and Brugada; Brugada score at the 22nd percentile). Atrial fibrillation remains at very low genetic risk (the 2nd percentile; TBX5 CC without the common risk allele). The variability of the genetic heart rate is average (the 49th percentile), with two unfavourable loci (RGS6 TT, CPNE5 GG) compensated by the favourable GNG11. KCNE1 D85N, the drug-induced long QT marker, was not read.

Favourable

- **Genetically low resting heart rate:** 80 score variants at the 4th percentile (the 243 variant at the 32nd percentile). Margin for the increase of frequency that retatrutide usually causes.
- **Favourable driving:** SCN10A rs6795970 GG and rs10428132 GG (without the broad QRS alleles and Brugada syndrome); Brugada score at the 22nd percentile; QRS at the 41st percentile; PR in the 26–37 percentiles.
- **Atrial fibrillation with very low genetic risk:** TBX5 rs883079 CC (without the T allele, which ~72% of the reference population alleles have), KCNJ5 GG; atrial fibrillation score at the 2nd percentile (cardiovascular report). GNG11 rs180238 TT (without the allele of lower variability).

Unfavourable

- **RGS6 rs4899412 TT and CPNE5/PPIL1 rs236349 GG:** two loci with lower heart rate variability (SDNN); score RMSSD at the 49th percentile (medium).
- **HCN4 rs4489968 TT:** homozygous for the highest frequency common allele, and HCN4 rs2680344 AG (lower variability); the global effect on frequency is compensated by the other loci (score at the 4th percentile).
- **KCNE1 D85N not read** (nor in the chip or imputation): it is not possible to exclude the main common genetic factor of long QT induced by drugs. HEY2 TC and CDKN1A GA (typical).

Interpretation: Favourable electrical profile: low genetic frequency, normal conduction and very low risk of atrial fibrillation. Heart rate variability is the most modifiable and useful part to follow: it responds to training, sleep and slow breathing.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 17 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

Heart rate scores (80 and 243 variants), PR (44 and 582), QRS (401) and Brugada (21) were calculated in this report. The RMSSD score uses 6 loci of a heart rate variability GWAS and recovery after exercise 8 loci (GWAS Catalog); there is no HRV score published in the PGS Catalog. Atrial fibrillation scores and QT were reused from the advanced cardiovascular report.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Mean heart rate (243 variants)	PGS001375	202 of 243 (83%)	p32	Typical / no judgement
Heart rate (80 variants)	PGS000300	77 of 80 (96%)	p4	Favourable
Heart rate variability, RMSSD (GWAS Catalog score, 6 loci)	C_HRV_RMSSD	5 of 6 (83%)	p49	Typical / no judgement
Frequency recovery after exercise (GWAS Catalog score, 8 loci)	C_FC_RECUP	8 of 8 (100%)	p91	Typical / no judgement
PR interval (582 variants)	PGS000904	527 of 582 (90%)	p26	Typical / no judgement
PR interval (44 variants)	PGS000735	44 of 44 (100%)	p37	Typical / no judgement
Duration of QRS (401 variants)	PGS001525	321 of 401 (80%)	p41	Typical / no judgement
Brugada syndrome (21 variants)	PGS001779	21 of 21 (100%)	p22	Typical / no judgement
Atrial fibrillation (166 variants)	PGS004920	163 of 166 (98%)	p2	Favourable
QT interval (68 variants)	PGS000768	65 of 68 (96%)	p36	Typical / no judgement

Resting heart rate and sinus node

The resting frequency is about 25% hereditary. Each 10 beat per minute is associated with more cardiovascular mortality in large cohorts; the low frequency per training is a favourable sign.

Polygenic scores of this theme: Mean heart rate (243 variants) (PGS001375): the 32nd percentile

Typical / no judgement; Heart rate (80 variants) (PGS000300): the 4th percentile **Favourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
HCN4	rs4489968	TT	chip	T	2	0.84	HCN4, the funny channel (IF) that generates the sinus node rhythm (target of ivabradine): the T allele increases the resting heart rate. (moderate evidence)	Typical / no judgement
SYT10	rs7980799	AC	chip	A	1	0.38	SYT10: Allele A increases the resting frequency and reduces variability (RMSSD). (moderate evidence)	Typical / no judgement
KIAA1755	rs6123471	TC	chip	T	1	0.57	KIAA1755: T allele increases the resting frequency and reduces variability. (moderate evidence)	Typical / no judgement
MYH6 Ala1101Val	rs365990	AG	chip	—	—	—	Myosin of the auricula (alpha): allele A is associated with lower heart rate; rare variants of MYH6 cause sinus node disease. (moderate evidence)	Typical / no judgement

Favourable

- Frequency score (80 variants) at the 4th percentile
- 243 variant score at the 32nd percentile

Unfavourable

- HCN4 rs4489968 TT (highest frequency common allele)
- SYT10 AC and KIAA1755 TC (typical)

Interpretation: Genetically low resting frequency. MYH6 Ala1101Val AG is information (rare variants of MYH6 cause sinus node disease). Drugs in use: retatrutide increases the frequency in some beats per minute; tadalafil causes a small reflex increase; lithium at a dose of 5 mg has no relevant effect on the sinus node.

Heart rate variability (vagal tone)

The variability of heart rate (differences between successive beats) measures the vagal tone. RMSSD and high frequency power reflect mainly the vagal tone; SDNN reflects the total variability. GWAS of HRV (about 50 000 people) found loci in the sinus node (HCN4) and in the muscarinic pathway (RGS6, GNG11).

Polygenic scores of this theme: Heart rate variability, RMSSD (GWAS Catalog score, 6 loci) (C_HRV_RMSSD): the 49th percentile **Typical / no judgement**; Frequency recovery after exercise (GWAS Catalog score, 8 loci) (C_FC_RECUP): the 91st percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
RGS6	rs4899412	TT	chip	T	2	0.28	RGS6 regulates the signal of the muscarinic receptor M2 in the sinus node (the vagal brake): the T allele reduces heart rate variability (SDNN). (moderate evidence)	Unfavourable
GNG11	rs180238	TT	chip	C	0	0.32	GNG11, subunit of the G protein of the vagus-activated GIRK channel: C allele reduces RMSSD and recovery of frequency after exercise. (moderate evidence)	Favourable
HCN4 sign of variability	rs2680344	AG	chip	A	1	0.75	Second HCN4 signal: A allele (common) reduces SDNN and RMSSD. (moderate evidence)	Typical / no judgement
NDUFA11	rs12974991	GG	chip	A	0	0.10	NDUFA11 (mitochondrial complex I): the A allele, less common, reduces the variability of high frequency (sinus respiratory arrhythmia). (moderate evidence)	Typical / no judgement
CPNE5/ PPIL1	rs236349	GG	chip	G	2	0.65	CPNE5/PPIL1: G allele increases the pulse and reduces SDNN. (weak evidence)	Unfavourable

Favourable

- GNG11 rs180238 TT (without the smallest C allele RMSSD)
- NDUFA11 rs12974991 GG (without allele A rare)

Unfavourable

- RGS6 rs4899412 TT (2 alleles of lower SDNN)
- CPNE5/PPIL1 rs236349 GG
- HCN4 rs2680344 AG
- RMSSD score (6 loci) at the 49th percentile

Interpretation: Mean genetic variability. The RGS6 accelerates the disconnection of the vagal signal in the sinus node; with the T allele in homozygous, the genetic vagal brake is a little weaker. The frequency recovery score after exercise is at the 91st percentile (trend for faster recovery), with the exception that loci have different effects. Real HRV depends much more on aerobic training, sleep, alcohol and stress than on genes.

Electrical conduction: PR, QRS and Brugada syndrome

The PR interval measures the conduction of the auricles to the ventricles; QRS, the conduction in the ventricles. Brugada syndrome is rare (about 1 in 2000), but common loci SCN10A and HEY2 explain part of the risk.

Polygenic scores of this theme: PR interval (582 variants) (PGS000904): the 26th percentile **Typical / no judgement**; PR interval (44 variants) (PGS000735): the 37th percentile **Typical / no judgement**; Duration of QRS (401 variants) (PGS001525): the 41st percentile **Typical / no judgement**; Brugada syndrome (21 variants) (PGS001779): the 22nd percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SCN10A Ala1073Val	rs6795970	GG	chip	A	0	0.41	SCN10A (sodium channel Nav1.8, regulates SCN5A): the Allele A extends QRS and PR (slowest conduction). (moderate evidence)	Favourable
SCN10A Brugada sign	rs10428132	GG	chip	T	0	0.42	Stronger common locus of Brugada syndrome (OR ~2.5 by T allele); only has clinical relevance with altered ECG or family history, because the syndrome is rare. (strong evidence)	Favourable
HEY2	rs9388451	TC	chip	C	1	0.49	HEY2 (transmural gradient of potassium channels): C allele increases the risk of Brugada (OR ~1.6). (moderate evidence)	Typical / no judgement
CDKN1A	rs9470361	GA	chip	A	1	0.22	CDKN1A (p21): Allele A extends QRS. (moderate evidence)	Typical / no judgement
CAV1	rs3807989	AA	chip	—	—	—	Caveolina-1: A allele elongates PR but reduces atrial fibrillation (effect in opposite directions). (moderate evidence)	Typical / no judgement
TBX5	rs883079	CC	chip	T	0	0.72	TBX5, transcription factor of the conduction system: the T allele (common) increases atrial fibrillation (OR ~1.1). (moderate evidence)	Favourable

Favourable

- SCN10A rs6795970 GG and rs10428132 GG
- TBX5 rs883079 CC
- Brugated at the 22nd percentile; QRS at the 41st percentile

Unfavourable

- HEY2 rs9388451 TC (typical)
- CDKN1A rs9470361 GA

Interpretation: Normal to favourable genetic conduction. CAV1 AA slightly extends PR but protects from atrial fibrillation (information). No genetic signs of Brugada.

Drug-induced QT, atrial fibrillation and sudden death

The long QT induced by drugs (macrolides, fluoroquinolones, some antidepressants and antiemetics) has the common genetic factor KCNE1 D85N. Atrial fibrillation and QT were seen in the advanced cardiovascular report (scores in the 2 and the 36th percentile).

Polygenic scores of this theme: Atrial fibrillation (166 variants) (PGS004920): the 2nd percentile Favourable; QT interval (68 variants) (PGS000768): the 36th percentile Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
KCNE1 D85N	rs1805128		SNP not found in the data	T	—	—	IKs channel subunit: the T allele (N85, rare) elongates QT and is the main common genetic factor of drug-induced long QT. (strong evidence)	SNP not found in the data
KCNJ5	rs76097649	GG	chip	A	0	0.07	KCNJ5 (an GIRK4 channel activated by acetylcholine in the auricles): A allele, less common, increases atrial fibrillation. (moderate evidence)	Typical / no judgement
BAZ2B	rs4665058	CC	chip	A	0	0.01	BAZ2B: locus of sudden cardiac death of a small GWAS (A rare allele, OR ~1.9); not robustly replicated. (weak evidence)	Typical / no judgement

Favourable

- KCNJ5 rs76097649 GG
- BAZ2B rs4665058 CC
- Atrial fibrillation at the 2nd percentile; QT at the 36th percentile

Unfavourable

- KCNE1 D85N not read

Interpretation: Risk of very low atrial fibrillation and mean genetic QT. As KCNE1 D85N has not been read, caution with combinations of drugs that extend QT remains. The high dose of omega-3 was associated with a small increase in atrial fibrillation in trials; with the genetic risk at the 2nd percentile, the absolute effect is very small.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Intercellular Communication	RGS6 TT; CPNE5 GG; HCN4 rs2680344 AG	Genetic frequency p4; SCN10A and TBX5 favourable	Globally favourable.
Desynchronization of Biological Clocks	HRV genetic medium	—	HRV follows the circadian rhythm and sleep.
Oxidative Phosphorylation Decline	—	NDUFA11 GG	Neutral.
Vascular Aging	—	Atrial fibrillation p2	Favourable.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Follow HRV	Using a clock or belt that measures the RMSSD in the morning (or during sleep) and follows the weekly trend; a sustained descent signals excessive training, weak sleep or disease. <i>suggestion (not applied)</i>	RGS6 TT; HRV genetic medium
Slow breathing	10 minutes per day of breathing at ~6 cycles per minute (inspire 5 s, exhale 5 s): increases the variability of high frequency in an acute way and, with practice, sustainably. <i>suggestion (not applied)</i>	RGS6 TT; CPNE5 GG
Aerobic training	Aerobic base (zone 2) with 1-2 sessions of intervals per week: it is the intervention that increases the HRV and improves the recovery of frequency after effort. <i>suggestion (not applied)</i>	Vagal tonus
Drugs that extend QT	As KCNE1 D85N was not read, warning the doctor before combining drugs that extend QT (macrolides, fluoroquinolones, ondansetron, some antidepressants and antifungals) and maintain normal potassium and magnesium. <i>suggestion (not applied)</i>	KCNE1 not read
Base ECG	A ECG of 12 base derivations, to record QTc, PR and QRS before any new drug that moves at the pace. <i>suggestion (not applied)</i>	Individual reference
Retatrutide and frequency	If the resting frequency rises in a way maintained with the retatrutide, comment with the doctor; the genetics gives margin (the 4th percentile). <i>suggestion (not applied)</i>	Low genetic frequency
Coordination with the plan	7-9 h and melatonin 0.3-1 mg. Omega-3 3.2-4 g/day (already taking fish oil), with a very low genetic risk of atrial fibrillation. D3 10 000 IU. Alcohol-free (alcohol is the major suppressors of HRV and triggers atrial fibrillation). <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- The rare variants of the channelopathies (SCN5A, KCNQ1, KCNH2, RYR2) and arrhythmogenic cardiomyopathies are not read reliably by the chip.
- KCNE1 D85N was not read or imputed with sufficient quality.
- The HRV and recovery scores use few loci; the real HRV is mainly environmental.
- The percentiles compare with the reference population (n=633). Only genes: does not use ECG or HRV records.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ($n=633$); effect alleles confirmed by the expected frequency.
- Scores: PGS000300 and PGS001375 (heart rate), PGS000735 and PGS000904 (PR), PGS001525 (QRS) and PGS001779 (Brugada) calculated in this report. PGS004920 and PGS000768 reused from the advanced cardiovascular report; simple scores of GWAS Catalog (local copy) for RMSSD and frequency recovery after exercise.
- Main references: Nolte 2017 (GWAS of heart rate variability); Eppinga 2016 (GWAS of the resting frequency); Ramirez 2018 (frequency during and after exercise); Bezzina 2013 (SCN10A and HEY2 in Brugada syndrome); van Setten 2018 (PR interval); Roden 2004 (QT induced by drugs); Lehrer 2014 (slow breathing and HRV); Gencer 2021 (omega-3 and atrial fibrillation, meta-analysis).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).